

Platinum Resistance Thermometer Sensors (RTD)

The Platinum Resistance Thermometer Sensor (RTD) is a temperature sensor containing a resistive temperature sensitive element. The resistance of the element changes with the temperature. For measuring within a range of -50 to 400 Deg C, platinum sensors are the most commonly used RTD sensors. Sensor elements made of platinum are normally produced for 100 OHM at 0 Deg C, described as PT100.

RTDs are amongst the most precise temperature sensors available. The resistance temperature detector senses heat, based on the principle that change in temperature results in a corresponding change in the resistance of the wire. The hotter the wires become, the larger and higher the value of their electrical resistance.



Special Features

Compliance to IEC-60751

Temperature range -50 to + 400 Deg C

High insulation resistance (>100 m OHM @ 500 VDC at 250 Deg C)

Mineral insulated construction customised to suit the installation without affecting performance

Available in a variety of sheath materials and diameters

Spring loaded design for positive contact with thermowell

Suitable for head mounted transmitters

Industrial Applications

Chemical and fertiliser industries

Petrochemical industry

Thermal, nuclear and hydroelectric power stations

Pharmaceutical industry

Food processing industry

Water treatment plants

Iron and steel industry

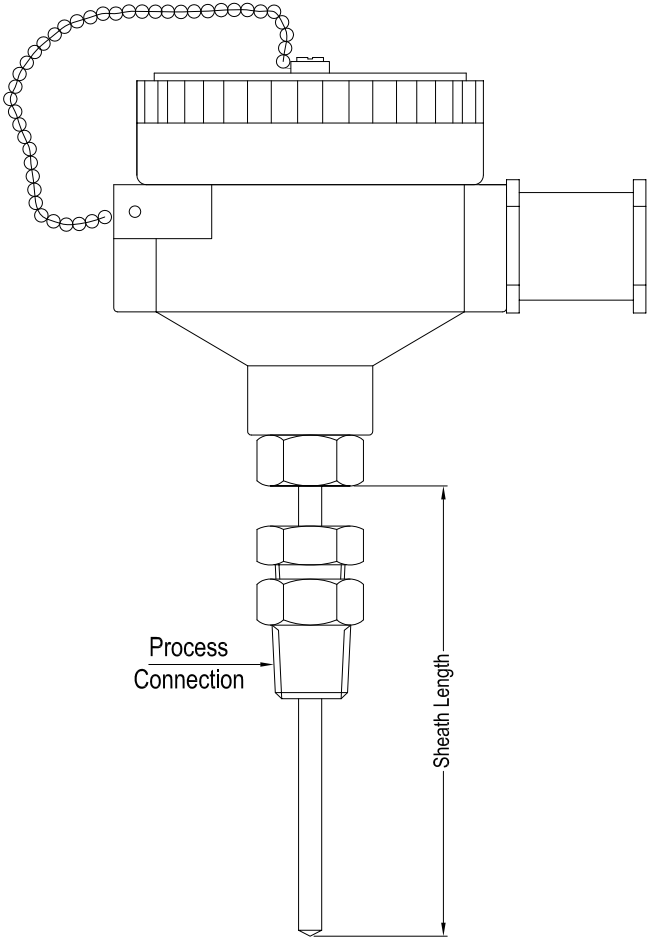
Rubber moulding / processing plants

Sugar industry

Refineries

Standard Product Specifications	
Specifications	Description
Reference standard	IEC-60751
Model	Simplex
Element	1 x PT 100, film type
Wire configuration	3- wire
Connection	1/2" NPT(M)
Sheath material	SS 316
Sheath type	Mineral insulated
Sheath diameter	6 mm
Sheath length	200 mm
Accuracy	Class A as per IEC -60751
Terminal head	Screwed type die cast aluminium weatherproof – IP 65
Head extension type	Adjustable connection
Cable gland	3/4" ET single compression gland
Cable gland material of construction	Brass nickle plated with rubber
Nominal resistance	100 OHM at 0 Deg. C
Ambient temperature	-25°C to 65°C
Temperature range	-50 to +400°C

MI Cable RTD Assembly



Ordering Information

Model	
Resistance temperature detector(duplex)	RD
Resistance temperature detector(simplesx)	RS

Element	
Pt100, film type	1

Wire Configuration	
3 wire	3
4 wire	4

Head Extension Type	
Adjustable connection	A
Nipple connection	B
Nipple union connection	D
Fixed connection (standard)	F
Flanged connection	E

Connection Size	
M18 X 1P	0
M20 X 1.5P	1
1/4"	A
3/8"	B
1/2"	C
3/4"	D
Not required	N

Connection type	
NPT(M)	A
BSP(M)	B
BSPT(M)	C
Metric thread(M)	M
#150-RF	4
#300-RF	5
Without connection	N

Cable Gland	
1/2" NPT, Double compression(1 No)	A
3/4" ET Double compression(1 No)	C
3/4" ET, Single compression(1 No) (standard)	D
1/2" NPT, Double compression(2 No)	E
3/4" ET, Double compression(2 No)	G
3/4" ET, Single compression(2 No)	H

Configuration Generation code

Sheath Diameter
6 mm (standard)
8 mm
10 mm

Sheath Length
As per requirement

Sheath Material of Construction
SS 316 (standard)
Inconel 600
Inconel 625
SS 321
SS 446

Head Extension Material of Construction
SS 316 (standard)
Carbon Steel
SS 304
SS 316L
Monel 400

Cable Gland Material of Construction
Brass nickle plated with rubber

Optional Features	
Accuracy class B as per IEC 60751	
Head protection	IP 67 protected head
	Ex. proof GR II C
	Flame proof GR II A, II B
Lead wire extension	PTFE braided
	SS braided
Lead wire length	1 mtr to 15 mtrs
Flange type	Blind
	Slip on
	Weld neck
Flange finish	125 AARH
	63 AARH
	250 AARH

Tests and Certificates	
Five point calibration certificate	
Material test certificate	
Third party inspection	
Special requirements – please specify	

Ordering example: RS13FNND-XXXX (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

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